# Question Paper

Exam Date & Time: 20-Nov-2019 (10:00 AM - 01:00 PM)

## CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

**University Examination November 2019**
**MSc.(IT)Semester-I**
**MS 119 - Operating System Concepts**
**Date:20-Nov-2019, Wednesday Time:10:00 a.m. to 01:00 p.m. Maximum Marks: 70**

**OPERATING SYSTEM CONCEPTS [MS119]**

**Marks: 70** **Duration: 180 mins.**

**Section-I**

**Answer all the questions.** Section Duration: 30 mins

**Make suitable assumption when ever necessary.**

1) Multiprogramming of computer system increases (1)

1) Memory Storage CPU Utilization Cost of Computation

2) User view of system depends upon the (1)

CPU Software Hardware Interface

3) Kernel mode of operating system is also known as (1)

User mode System mode Supervisor mode Data mode

4) The number of processes completed per unit time is known as ___________ (1)

Output Throughput Efficiency Capacity

5) A____________process is moved to the ready state when its time allocation expires (1)

New Blocked Running Suspended

6) First-in-First-Out (FIFO) scheduling is (1)

Non Pre-emptive Scheduling | Preemptive Scheduling | Fair Share Scheduling | Deadline Scheduling

7) The degree of Multiprogramming is controlled by___________ (1)

CPU scheduler | Context Switching | Long term scheduler | Medium term scheduler

8) In process scheduling,___________determines which ready process will be executed next by processor (1)

Long term scheduling | Medium term scheduling | Short term scheduling | None of the above

9) _______is a high level abstraction over Semaphore (1)

Shared memory | Message passing | Monitor | Mutual exclusion

10) The Banker's algorithm is used (1)

To rectify deadlock To detect deadlock To prevent deadlock To solve deadlock

11) _________is the time required to move the disk arm to the required track (1)

Seek Time Rotational delay Latency time Access time

12) In kernel model, the operating system services such as process management, memory management are provided by the kernel. (1)

Monolithic Micro Macro Complex

13) Round robin scheduling is essentially the preemptive version of _________ (1)

FIFO Shortest job first Shortest remaining Longest time first

14) If a process fails, most operating system write the error information to a (1)

Log file Another running process New file None of these

15) The mechanism that bring a page into memory only when it is needed is called (1)

Segmentation Fragmentation Demand Paging Page Replacement

16) The simplest directory structure is (1)

Single level directory Two level directory Tree structure directory None of these

17) Moving process from main memory to disk is called (1)

Scheduling Catching Swapping Spooling

18) Which of the following memory unit that processor can access more rapidly (1)

Main Memory Virtual Memory Cache memory Read Only Memory

19) Main memory of computer system is known to be (1)

Non volatile Volatile Reserved Restricted

20) Operating systems that allow different parts of a software program to run concurrently? (1)

Multiprocessing. Multithreading. Multitasking. Multiuser.

## Section-II

**Answer all the questions.**

**Make suitable assumption when ever necessary.**

2) A) What are schedulers? Explain the types of schedulers. (3)

[OR] B) What is a process? Draw and explain process state diagrams. (3)

C) Given below are the burst times of three processes P1, P2 and P3. Draw the Gantt Chart using RR scheduling and calculate the average waiting time, average turn around time. Quantum time = 5 ms (4)

| Process | Burst Time |
|---|---|
| P1 | 30 |
| P2 | 6 |
| P3 | 8 |

[OR] D) What are system calls? Explain different categories of system calls with example? (4)

3) A) What is process scheduling? Explain the criteria of scheduling. (6)

[OR] B) Explain SJF scheduling with example. (6)

C) Explain the services provided by Operating System. (6)

[OR] D) What are the necessary condition for deadlock? (6)

E) What is Demand Paging? Explain with diagram. (6)

[OR] F) What are the information associated with device directory? (6)

**Section-III**

**Answer all the questions.**

**Make suitable assumption when ever necessary.**

4) A) Define Operating Systems and functions of operating system. (3)

[OR] B) What are basic approach for page replacement? (3)

C) What are contiguous and non -contiguous memory allocation? (4)

[OR] D) What are the advantages & disadvantages of Segmentation? (4)

5) A) What are the advantages & disadvantages of Paging? (6)

[OR] B) What are the operations performed on directory? (6)

C) What is Page Fault? State the steps involved in handling a page fault. (6)

[OR] D) Explain about deadlock Avoidance in detail. (6)

E) Explain FIFO page replacement algorithms. (6)

[OR] F) Describe various file access methods. (6)

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